Zipf's Law

this Suppose we take a set of data and rank it by size. For example, we might take the set of cities in the US, and rank them by population. New York City would be ranked number one, LA number two, and so on. Durham is ranked 79. Zipf's Law says that the values of a data point is approximately inversely proportional to a power of its rank. In the case above, it says that population P of a city is inversely proportional to some power of its population rank r. That is, $P \approx \frac{k}{r^a}$, where k and a are constants. The constant a is called the *Zipf constant* of the data set. Note that this is a power function!

Your project is to produce a report on Zipf's Law in the context of population rank of cities by examining data sets of the populations of cities/metropolitan areas in the US, China, England, and Brazil. These data sets are available as a Google Sheet, linked here.

Your first job is to analyze the populations and ranks of cities in each of these countries using log-log plots (since we're expecting power functions), and model them using Zipf's Law. As part of that, you will determine the values of Zipf's constant for each of the countries, as well as the constant of proportionality.

Once you've done that, you will examine the fit of your model for each country, as well as look for cities whose populations are substantially higher or lower than the model predicts. You will then analyze patterns of socioeconomic growth in each country to attempt to explain what you observe.

Your report should include:

- an overview of Zipf's Law, its historical origin, and its later applications;
- a simple example of a case where Zipf's Law can be explained straightforwardly (there are a number in this video);
- The application of Zipf's Law to the data set:
  - Description of the data;
  - Well labeled and titled log-log plots for each of the four countries, with linear regression lines displayed;
  - Formulas (deduced from the linear regressions) for the size of a city given its rank in the form $P = \frac{k}{r^a}$. For help, see the log-log plots part of the lab;
  - A brief discussion of which countries the model fits better for, and which less so.
  - For each country, look at your log-log plot and use it to answer: Are there cities that have a lower or higher population than what you would expect from the model? Explain this within the context of the country you are looking at. Specifically, research how cities developed in each of the countries concerned, in terms of planned vs natural population growth and spread, economic centers, and so on. This is the central part of the report, so spend time and space on it.

Your report should *not* include entire datasets, as they are large. You should include a sample of the data, though. You do not need to cite your data sources, but (as mentioned above), you should cite any other material. Note that you should most definitely make use of the entire data set in carrying out your analysis, but please do not include the entire thing in your final report!

A word of caution: please do not include any items in your report which you do not understand thoroughly. You should cite any resources you use, but do not blindly copy *any* text, formulas, or mathematical detail[1]. Any formulas you use should be clearly explained. Any terminology you use that has not been covered in class should be clearly explained.

If you need help with spreadsheets, charts, or understanding mathematical detail, please make an appointment to come talk to me. I would be happy to help guide your thinking, and to take a look at work you have done.

[1] That would constitute plagirism, and will be treated as such!